Python Logical Operators – Important Notes

Definition: Logical operators are used to combine multiple conditions.

Operator	Meaning	Example
and	True when both conditions are True	a > 5 and b < 10
or	True when at least one condition is True	a > 5 or b > 10
not	Reverses True/False	not(a > 5)

6 Important Examples

Example 1: and – both conditions

```
age = 25
print(age >= 18 and age <= 60)
```

Output: True

Example 2: and – two conditions

```
a = 10
b = 20
print(a > 5 and b > 15)
```

Output: True

Example 3: or – one condition

```
a = 10
b = 3
print(a > 20 or b < 5)
```

Output: True

Example 4: or – both false

```
marks = 40
print(marks >= 75 or marks < 35)
```

Output: False

Example 5: not

```
x = True
print(not x)
```

Output: False

Example 6: Combined

```
age = 20
marks = 80
print(age >= 18 and marks >= 75)
```

Output: True